

Assignment 4 — Dataset 1: Diabetes 130-US Hospitals (tabular)

One learning problem, three implementations: NumPy from scratch, TensorFlow/Keras, and PyTorch.

This notebook covers the first of the three datasets required by the assignment:

#	Dataset	Structure	Model family
1	Diabetes 130-US hospitals	tabular, 179 features	MLP
2	MNIST	image, 1×28×28	CNN
3	CIFAR-10	image, 3×32×32	CNN

Why an MLP and not a CNN here

Lecture 04, slide 4 makes the distinction that motivates the whole assignment. A tabular row

$$x = [x_1, x_2, \dots, x_d]$$

has no meaningful two-dimensional neighbourhood: feature x_i is not spatially adjacent to x_{i+1} the way neighbouring pixels are. Convolution exists to exploit *local spatial structure*; where there is none, its two defining ideas — local connectivity and weight sharing — buy nothing. So:

Diabetes → MLP

MNIST / CIFAR-10 → CNN

The model should match the structure of the data. That sentence is the reason this dataset is in the assignment at all: it is the control case that shows a CNN is a choice, not a default.

What stays constant across the three implementations

Slide 29 sets the fairness rule:

Same Dataset + Same Split + Same Architecture + Comparable Hyperparameters

All three legs below pull their arrays from the same `ass4_utils` loader with the same seed, so they see byte-identical training and test sets. Only the *level of abstraction* changes.

```
In [1]: import os, sys, json, time, platform

# TensorFlow is CPU-only on Windows (no GPU support since 2.10); silence its
# startup chatter.
os.environ.setdefault("TF_CPP_MIN_LOG_LEVEL", "2")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.abspath("."))
import ass4_utils as U
import scratch_nn as S

U.set_seed(U.SEED)

import torch
import tensorflow as tf
import keras

print("python      ", platform.python_version())
print("numpy      ", np.__version__)
print("torch      ", torch.__version__, "| CUDA:", torch.cuda.is_available(),
      "|", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "CPU
only")
print("tensorflow ", tf.__version__, "| GPUs:",
      tf.config.list_physical_devices("GPU") or "none (CPU)")
print("keras      ", keras.__version__)

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
plt.rcParams["figure.dpi"] = 110
```

```
python      3.11.8
numpy      2.2.6
torch      2.10.0+cu130 | CUDA: True | NVIDIA GeForce RTX 3060 Laptop GPU
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for
TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used.
Please use WSL2 or the TensorFlow-DirectML plugin.
```

```
tensorflow 2.21.0 | GPUs: none (CPU)
keras      3.15.1
```

1. The data

The dataset is the UCI *Diabetes 130-US hospitals for years 1999–2008* release: ten years of inpatient records, **101,766 encounters × 50 columns**. It is downloaded automatically on first run.

The loader in `ass4_utils.py` applies the preprocessing that is standard for this dataset:

1. **One row per patient.** Repeat encounters of the same patient leak the outcome of earlier admissions into later ones, so only the first admission is kept.
2. **Drop expired/hospice discharges.** Those patients cannot be readmitted; scoring them as `NO` would teach the model a false pattern.
3. **Drop identifiers and near-empty columns** — `weight` (96.9 % missing), `payer_code` (39.6 %), `medical_specialty` (49.1 %), plus the two constant drug columns `examide` and `citoglipton`.
4. **Group the ICD-9 diagnosis codes.** `diag_1..3` hold ~700 distinct codes each; they are collapsed into nine clinical groups (circulatory, respiratory, diabetes, ...) so one-hot encoding stays tractable.
5. **One-hot the categoricals, standardise the numerics** using *training* statistics only.

The target is `readmitted` with three classes: `NO`, `>30` days, `<30` days.

```
In [2]: X_train, y_train, X_test, y_test, meta = U.load_diabetes130()

N_FEATURES = X_train.shape[1]
CLASSES = meta["classes"]
N_CLASSES = len(CLASSES)

print(f"\ntrain {X_train.shape}    test {X_test.shape}    dtype {X_train.dtype}")
print(f"features: {N_FEATURES}    classes: {CLASSES}")
```

Diabetes 130-US hospitals: raw (101766, 50) -> (69973, 179), train 55979 / test 13994, 179 features, 3 classes

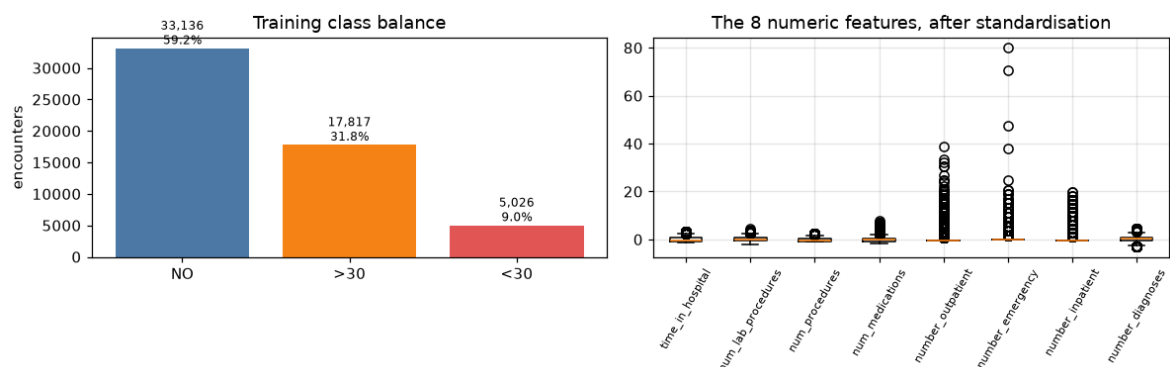
```
train (55979, 179)    test (13994, 179)    dtype float32
features: 179    classes: ['NO', '>30', '<30']
```

```
In [3]: # Class balance and a look at the feature block structure.
fig, ax = plt.subplots(1, 2, figsize=(11, 3.6))

counts = np.bincount(y_train, minlength=N_CLASSES)
bars = ax[0].bar(CLASSES, counts, color=["#4C78A8", "#F58518", "#E45756"])
ax[0].set_title("Training class balance")
ax[0].set_ylabel("encounters")
for b, c in zip(bars, counts):
    ax[0].text(b.get_x() + b.get_width() / 2, c, f"
{c:,}\n{c/counts.sum():.1%}",
              ha="center", va="bottom", fontsize=8)

numeric_names = [n for n in meta["feature_names"][:8]]
ax[1].boxplot([X_train[:, i] for i in range(8)], tick_labels=numeric_names)
ax[1].set_title("The 8 numeric features, after standardisation")
ax[1].tick_params(axis="x", rotation=60, labels=7)
ax[1].grid(alpha=0.3)
fig.tight_layout()
plt.show()

print(f"{'class':<6}{ 'train':>10}{ 'share':>9}")
for i, c in enumerate(CLASSES):
    print(f"{'c':<6}{counts[i]:>10},{counts[i]/counts.sum():>9.1%}")
```



class	train	share
NO	33,136	59.2%
>30	17,817	31.8%
<30	5,026	9.0%

The imbalance matters for how we read the results later. Predicting NO for every single patient already scores ~59 % accuracy while being clinically useless. That is why the comparison tables report **macro F1** alongside accuracy: macro F1 averages over the three classes equally, so a model that ignores the rare <30 class cannot hide behind the majority class.

2. The shared architecture

Every implementation below builds *this* function, and nothing else:

$X \rightarrow \text{Linear}_{179 \rightarrow 128} \rightarrow \text{ReLU} \rightarrow \text{Dropout}_{0.3} \rightarrow \text{Linear}_{128 \rightarrow 64} \rightarrow \text{ReLU} \rightarrow \text{Line}$

which is the tutorial's central claim made concrete — a network *is* a composition of functions:

$$F = f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1$$

Counting the parameters by hand first

Tutorial §43 gives the rule for a linear layer:

$$N_{\text{params}} = N_{\text{in}}N_{\text{out}} + N_{\text{out}}$$

Doing the arithmetic *before* asking any framework is the point of the exercise — each implementation is then checked against this number, and all three must agree.

```
In [4]: HIDDEN1, HIDDEN2, DROPOUT = 128, 64, 0.3
        EPOCHS, BATCH_SIZE, LR = 15, 256, 1e-3

        # Hand-computed, per tutorial section 43: N_in * N_out + N_out
        manual = [
            ("Linear 179->128", N_FEATURES * HIDDEN1 + HIDDEN1),
            ("ReLU", 0),
            ("Dropout 0.3", 0),
            ("Linear 128->64", HIDDEN1 * HIDDEN2 + HIDDEN2),
            ("ReLU", 0),
            ("Linear 64->3", HIDDEN2 * N_CLASSES + N_CLASSES),
        ]
        EXPECTED_PARAMS = sum(p for _, p in manual)

        print(f"{'layer':<20}{'params':>10}")
        print("-" * 30)
        for name, p in manual:
            print(f"{'name':<20}{'p':>10}")
            print("-" * 30)
        print(f"{'TOTAL (by hand)':<20}{'EXPECTED_PARAMS:>10}")
        print(f"\nshared hyperparameters: epochs={EPOCHS}, batch_size={BATCH_SIZE},
optimizer=Adam(lr={LR})")

        results = [] # collects one U.RunResult per implementation
```

layer	params
Linear 179->128	23,040
ReLU	0
Dropout 0.3	0
Linear 128->64	8,256
ReLU	0
Linear 64->3	195
TOTAL (by hand)	31,491

shared hyperparameters: epochs=15, batch_size=256, optimizer=Adam(lr=0.001)

3. Implementation A — from scratch (NumPy)

No framework. `scratch_nn.py` defines each layer as an explicit function with a hand-derived gradient, and the training loop spells out all four steps of the cycle:

Forward → Loss → Gradient → Update

Adam is written out as arithmetic rather than called from a library, so nothing about the update rule is hidden.

```
In [5]: U.set_seed(U.SEED)

scratch_model = S.Sequential([
    S.Dense(N_FEATURES, HIDDEN1, seed=1),
    S.ReLU(),
    S.Dropout(DROPOUT, seed=11),
    S.Dense(HIDDEN1, HIDDEN2, seed=2),
    S.ReLU(),
    S.Dense(HIDDEN2, N_CLASSES, seed=3),
])

print(scratch_model.summary(input_shape=(N_FEATURES,)))
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? "
      f"{scratch_model.n_params() == EXPECTED_PARAMS}")
```

layer	output shape	params
Dense(179 -> 128)	(128,)	23,040
ReLU	(128,)	0
Dropout(p=0.3)	(128,)	0
Dense(128 -> 64)	(64,)	8,256
ReLU	(64,)	0
Dense(64 -> 3)	(3,)	195
total		31,491

matches hand count 31,491? True

Are the hand-written gradients actually right?

Before trusting anything this model reports, the analytic gradients are checked against central finite differences,

$$\frac{\partial \mathcal{L}}{\partial \theta_i} \approx \frac{\mathcal{L}(\theta_i + \epsilon) - \mathcal{L}(\theta_i - \epsilon)}{2\epsilon}$$

on a small batch in float64. A relative error below $\sim 1\text{e-}4$ means the backward pass derived by hand agrees with the numerical derivative — the framework legs get this correctness for free from autodiff, so the scratch leg has to earn it.

```
In [6]: U.set_seed(U.SEED)
        check_model = S.Sequential([
            S.Dense(N_FEATURES, 16, seed=1),
            S.ReLU(),
            S.Dense(16, N_CLASSES, seed=3),
        ])
        err = S.gradient_check(check_model, X_train[:8], y_train[:8], n_samples=8)
        print(f"worst relative error vs finite differences: {err:.3e}")
        print("PASS – hand-derived gradients agree with the numerical derivative" if
              err < 1e-4
              else "FAIL – backward pass disagrees with finite differences")
```

worst relative error vs finite differences: 4.511e-07

PASS – hand-derived gradients agree with the numerical derivative

```
In [7]: U.set_seed(U.SEED)
        with U.Timer() as t_scratch:
            hist_scratch = S.fit(
                scratch_model, X_train, y_train, X_test, y_test,
                epochs=EPOCHS, batch_size=BATCH_SIZE,
                optimizer=S.Adam(LR), seed=U.SEED,
            )
        print(f"\ntotal training time: {t_scratch.seconds:.1f}s")
```

epoch 1/15 loss 0.9719 0.4s test_acc 0.6003

epoch 2/15 loss 0.8816 0.4s test_acc 0.6042

epoch 3/15 loss 0.8646 0.4s test_acc 0.6045

epoch 4/15 loss 0.8555 0.4s test_acc 0.6065

epoch 5/15 loss 0.8476 0.4s test_acc 0.6089

epoch 6/15 loss 0.8467 0.4s test_acc 0.6093

epoch 7/15 loss 0.8424 0.4s test_acc 0.6093

epoch 8/15 loss 0.8384 0.4s test_acc 0.6092

epoch 9/15 loss 0.8368 0.4s test_acc 0.6086

epoch 10/15 loss 0.8346 0.4s test_acc 0.6063

epoch 11/15 loss 0.8317 0.4s test_acc 0.6071

epoch 12/15 loss 0.8292 0.3s test_acc 0.6100

epoch 13/15 loss 0.8282 0.4s test_acc 0.6086

epoch 14/15 loss 0.8253 0.4s test_acc 0.6073

epoch 15/15 loss 0.8237 0.4s test_acc 0.6094

total training time: 6.0s

```
In [8]: pred = scratch_model.predict_classes(X_test)
m = U.evaluate(y_test, pred, N_CLASSES)

results.append(U.RunResult(
    framework="Scratch (NumPy)", dataset=meta["name"], model="MLP 179-128-64-3",
    n_params=scratch_model.n_params(), epochs=EPOCHS,
    train_seconds=round(t_scratch.seconds, 2),
    train_loss=round(hist_scratch["loss"][-1], 4),
    history=hist_scratch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.6094 macro-F1 0.3450

4. Implementation B — TensorFlow / Keras

The same composition, expressed as a list of reusable layers. Compare what disappeared from our code: the explicit loop over batches, the loss computation, the gradient and the parameter update are all now inside `fit()`.

Note on hardware. Windows-native TensorFlow has shipped without GPU support since version 2.10, so this leg runs on CPU while the PyTorch leg below uses the GPU. Training times are therefore **not** directly comparable across those two rows — the parameter counts and accuracies are.

```
In [9]: U.set_seed(U.SEED)
```

```
keras_model = keras.Sequential([
    keras.layers.Input(shape=(N_FEATURES,)),
    keras.layers.Dense(HIDDEN1, activation="relu"),
    keras.layers.Dropout(DROPOUT),
    keras.layers.Dense(HIDDEN2, activation="relu"),
    keras.layers.Dense(N_CLASSES),          # raw logits, like the other two
    ], name="diabetes_mlp")

keras_model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=LR),
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=["accuracy"],
)
keras_model.summary()
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? "
      f"{keras_model.count_params() == EXPECTED_PARAMS}")
```

Model: "diabetes_mlp"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 128)	23,040
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
dense_2 (Dense)	(None, 3)	195

Total params: 31,491 (123.01 KB)

Trainable params: 31,491 (123.01 KB)

Non-trainable params: 0 (0.00 B)

matches hand count 31,491? True

```
In [10]: U.set_seed(U.SEED)
with U.Timer() as t_keras:
    hk = keras_model.fit(
        X_train, y_train,
        validation_data=(X_test, y_test),
        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2,
    )
print(f"\ntotal training time: {t_keras.seconds:.1f}s")
```

Epoch 1/15

219/219 - 1s - 6ms/step - accuracy: 0.5829 - loss: 0.9071 - val_accuracy: 0.6032
- val_loss: 0.8665

Epoch 2/15

219/219 - 0s - 2ms/step - accuracy: 0.5995 - loss: 0.8653 - val_accuracy: 0.6078
- val_loss: 0.8586

Epoch 3/15

219/219 - 1s - 2ms/step - accuracy: 0.6072 - loss: 0.8531 - val_accuracy: 0.6066
- val_loss: 0.8552

Epoch 4/15

219/219 - 1s - 2ms/step - accuracy: 0.6097 - loss: 0.8479 - val_accuracy: 0.6085
- val_loss: 0.8529

Epoch 5/15

219/219 - 0s - 2ms/step - accuracy: 0.6113 - loss: 0.8415 - val_accuracy: 0.6060
- val_loss: 0.8523

Epoch 6/15

219/219 - 0s - 2ms/step - accuracy: 0.6138 - loss: 0.8383 - val_accuracy: 0.6095
- val_loss: 0.8519

Epoch 7/15

219/219 - 0s - 2ms/step - accuracy: 0.6146 - loss: 0.8362 - val_accuracy: 0.6096
- val_loss: 0.8516

Epoch 8/15

219/219 - 0s - 2ms/step - accuracy: 0.6171 - loss: 0.8330 - val_accuracy: 0.6090
- val_loss: 0.8521

Epoch 9/15

219/219 - 0s - 2ms/step - accuracy: 0.6161 - loss: 0.8296 - val_accuracy: 0.6090
- val_loss: 0.8526

Epoch 10/15

219/219 - 0s - 2ms/step - accuracy: 0.6187 - loss: 0.8291 - val_accuracy: 0.6098
- val_loss: 0.8520

Epoch 11/15

219/219 - 0s - 2ms/step - accuracy: 0.6206 - loss: 0.8252 - val_accuracy: 0.6104
- val_loss: 0.8529

Epoch 12/15

219/219 - 0s - 2ms/step - accuracy: 0.6211 - loss: 0.8236 - val_accuracy: 0.6116
- val_loss: 0.8519

Epoch 13/15

219/219 - 0s - 2ms/step - accuracy: 0.6218 - loss: 0.8215 - val_accuracy: 0.6065
- val_loss: 0.8528

Epoch 14/15

219/219 - 0s - 2ms/step - accuracy: 0.6222 - loss: 0.8193 - val_accuracy: 0.6074
- val_loss: 0.8542

Epoch 15/15

219/219 - 0s - 2ms/step - accuracy: 0.6249 - loss: 0.8174 - val_accuracy: 0.6077
- val_loss: 0.8549

total training time: 7.8s

```
In [11]: pred = keras_model.predict(X_test, batch_size=512, verbose=0).argmax(axis=1)
m = U.evaluate(y_test, pred, N_CLASSES)

hist_keras = {"loss": [float(v) for v in hk.history["loss"]],
              "val_acc": [float(v) for v in hk.history["val_accuracy"]]}

results.append(U.RunResult(
    framework="TensorFlow/Keras", dataset=meta["name"], model="MLP 179-128-64-3",
    n_params=int(keras_model.count_params()), epochs=EPOCHS,
    train_seconds=round(t_keras.seconds, 2),
    train_loss=round(hist_keras["loss"][-1], 4),
    history=hist_keras, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.6077 macro-F1 0.3686

5. Implementation C — PyTorch

The same composition again, as an `nn.Module`. PyTorch sits between the other two: the *model* is as compact as Keras, but the *training loop* is as explicit as the scratch version —

Zero Gradient → Forward → Loss → Backward → Update

is written out line by line. What PyTorch supplies that the scratch code had to derive by hand is `loss.backward()`: autograd.

```

In [12]: import torch.nn as nn
         from torch.utils.data import TensorDataset, DataLoader

         U.set_seed(U.SEED)

         class DiabetesMLP(nn.Module):
             def __init__(self, n_in, n_classes):
                 super().__init__()
                 self.net = nn.Sequential(
                     nn.Linear(n_in, HIDDEN1),
                     nn.ReLU(),
                     nn.Dropout(DROPOUT),
                     nn.Linear(HIDDEN1, HIDDEN2),
                     nn.ReLU(),
                     nn.Linear(HIDDEN2, n_classes),
                 )

             def forward(self, x):
                 return self.net(x)

         torch_model = DiabetesMLP(N_FEATURES, N_CLASSES).to(DEVICE)
         n_torch = sum(p.numel() for p in torch_model.parameters())

         print(torch_model)
         print(f"\nparameters: {n_torch:,}")
         print(f"matches hand count {EXPECTED_PARAMS:,}? {n_torch == EXPECTED_PARAMS}")

```

```

DiabetesMLP(
  (net): Sequential(
    (0): Linear(in_features=179, out_features=128, bias=True)
    (1): ReLU()
    (2): Dropout(p=0.3, inplace=False)
    (3): Linear(in_features=128, out_features=64, bias=True)
    (4): ReLU()
    (5): Linear(in_features=64, out_features=3, bias=True)
  )
)

parameters: 31,491
matches hand count 31,491? True

```

```

In [13]: train_ds = TensorDataset(torch.tensor(X_train), torch.tensor(y_train))
         train_loader = DataLoader(train_ds, batch_size=BATCH_SIZE, shuffle=True)

         Xte_t = torch.tensor(X_test).to(DEVICE)
         criterion = nn.CrossEntropyLoss()
         optimizer = torch.optim.Adam(torch_model.parameters(), lr=LR)

         hist_torch = {"loss": [], "val_acc": []}

         with U.Timer() as t_torch:
             for epoch in range(1, EPOCHS + 1):
                 torch_model.train()
                 running, nb = 0.0, 0
                 for xb, yb in train_loader:
                     xb, yb = xb.to(DEVICE), yb.to(DEVICE)

                     optimizer.zero_grad()          # 1. zero gradient
                     logits = torch_model(xb)        # 2. forward
                     loss = criterion(logits, yb)     # 3. loss
                     loss.backward()                 # 4. backward (autograd)
                     optimizer.step()                # 5. update

                     running += loss.item(); nb += 1

                 torch_model.eval()
                 with torch.no_grad():
                     acc = (torch_model(Xte_t).argmax(1).cpu().numpy() == y_test).mean()

                 hist_torch["loss"].append(running / nb)
                 hist_torch["val_acc"].append(float(acc))
                 print(f"epoch {epoch}/{EPOCHS} loss {running/nb:.4f} test_acc
{acc:.4f}")

         print(f"\ntotal training time: {t_torch.seconds:.1f}s")

```

epoch 1/15 loss 0.8733 test_acc 0.6093

epoch 2/15 loss 0.8493 test_acc 0.6103

epoch 3/15 loss 0.8421 test_acc 0.6095

epoch 4/15 loss 0.8402 test_acc 0.6083

epoch 5/15 loss 0.8363 test_acc 0.6105

epoch 6/15 loss 0.8334 test_acc 0.6109

epoch 7/15 loss 0.8306 test_acc 0.6084

epoch 8/15 loss 0.8286 test_acc 0.6107

epoch 9/15 loss 0.8255 test_acc 0.6043

epoch 10/15 loss 0.8240 test_acc 0.6086

epoch 11/15 loss 0.8201 test_acc 0.6110

epoch 12/15 loss 0.8191 test_acc 0.6108

epoch 13/15 loss 0.8161 test_acc 0.6095

epoch 14/15 loss 0.8155 test_acc 0.6121

epoch 15/15 loss 0.8135 test_acc 0.6105

total training time: 9.2s

```
In [14]: torch_model.eval()
with torch.no_grad():
    pred = torch_model(Xte_t).argmax(1).cpu().numpy()
    m = U.evaluate(y_test, pred, N_CLASSES)

results.append(U.RunResult(
    framework="PyTorch", dataset=meta["name"], model="MLP 179-128-64-3",
    n_params=n_torch, epochs=EPOCHS,
    train_seconds=round(t_torch.seconds, 2),
    train_loss=round(hist_torch["loss"][-1], 4),
    history=hist_torch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}   macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.6105 macro-F1 0.3694

6. Comparison

6.1 The numbers slide 29 asks for

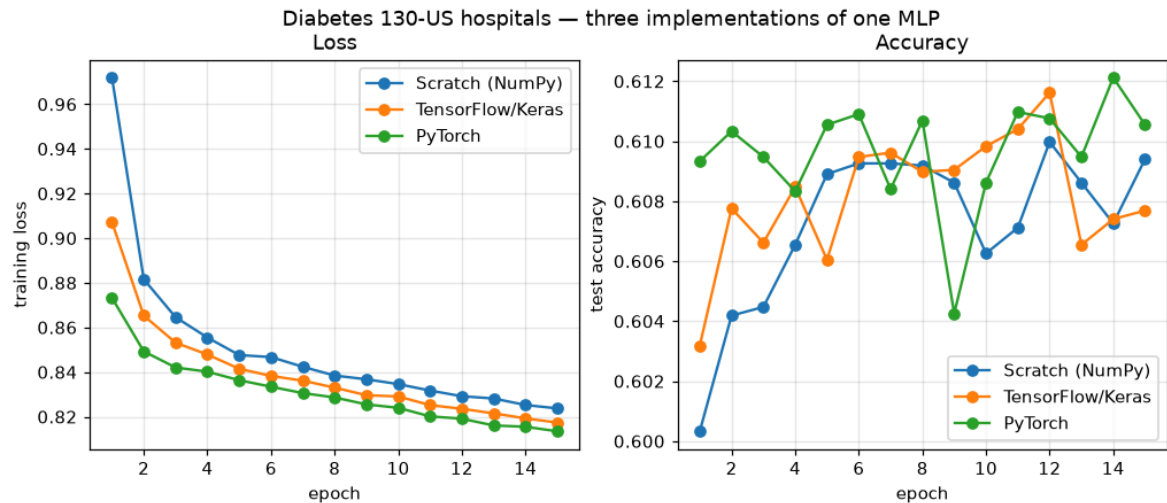
```
In [15]: table = U.results_table(results)
display(table)
```

	frame work	dataset	model	n_pa ram s	ep oc hs	train_ secon ds	trai n_lo ss	test_ accur acy	precisi on_ma cro	recall _mac ro	f1_ mac ro
0	Scratch (NumPy)	Diabetes 130-US hospitals	MLP 179- 128-64- 3	3149 1	15	6.04	0.82 37	0.609 4	0.4798	0.372 8	0.34 50
1	TensorFlow/Keras	Diabetes 130-US hospitals	MLP 179- 128-64- 3	3149 1	15	7.78	0.81 74	0.607 7	0.4839	0.384 2	0.36 86
2	PyTorch	Diabetes 130-US hospitals	MLP 179- 128-64- 3	3149 1	15	9.17	0.81 35	0.610 5	0.5129	0.386 0	0.36 94

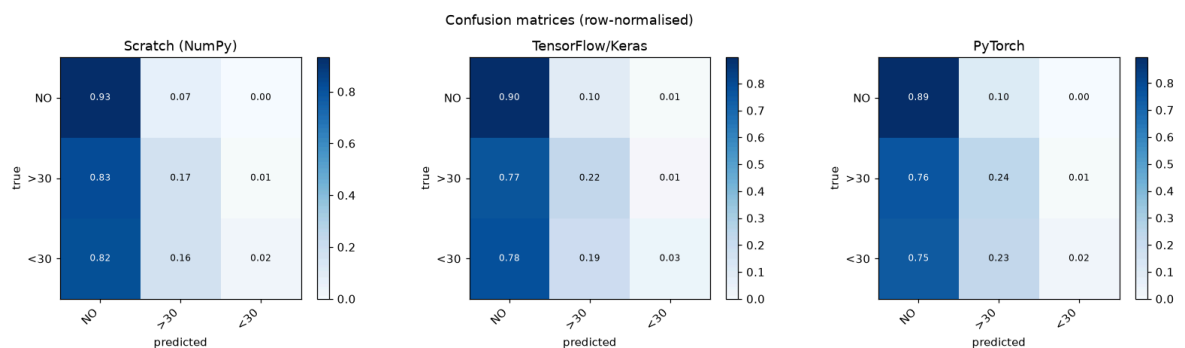
```
In [16]: # A majority-class baseline, so the accuracies above can be read honestly.
majority = np.bincount(y_train).argmax()
base = U.evaluate(y_test, np.full_like(y_test, majority), N_CLASSES)
print(f"always-predict-'{CLASSES[majority]}' baseline: "
      f"accuracy {base['test_accuracy']:.4f}    macro-F1 "
      f"{base['f1_macro']:.4f}")
print(f"best model: "
      f"accuracy {table['test_accuracy'].max():.4f}    macro-F1 "
      f"{table['f1_macro'].max():.4f}")
```

```
always-predict-'NO' baseline: accuracy 0.5958    macro-F1 0.2489
best model:                accuracy 0.6105    macro-F1 0.3694
```

```
In [17]: U.plot_history({r.framework: r.history for r in results},
                        title=f"{meta['name']} — three implementations of one MLP")
plt.show()
```



```
In [18]: fig, axes = plt.subplots(1, 3, figsize=(15, 4.2))
for ax, r in zip(axes, results):
    U.plot_confusion(r.confusion, CLASSES, title=r.framework, ax=ax)
fig.suptitle("Confusion matrices (row-normalised)")
fig.tight_layout()
plt.show()
```



6.2 Where each component lives

This is the table slide 28 asks students to submit, filled in for this notebook.

```

In [19]: component_table = pd.DataFrame([
    ["Data loading",      "ass4_utils -> NumPy arrays", "same arrays",
    "same arrays"],
    ["Preprocessing",    "ass4_utils (shared)",        "shared",
    "shared"],
    ["Model definition", "S.Sequential(...)",
    "keras.Sequential(...)", "nn.Module + nn.Sequential"],
    ["Dense layer",      "S.Dense (x@W+b by hand)",    "keras.layers.Dense",
    "nn.Linear"],
    ["Activation",       "S.ReLU (mask multiply)",     "activation='relu'",
    "nn.ReLU"],
    ["Regularisation",   "S.Dropout (inverted)",       "keras.layers.Dropout",
    "nn.Dropout"],
    ["Loss",             "S.SoftmaxCrossEntropy",
    "SparseCategoricalCrossentropy", "nn.CrossEntropyLoss"],
    ["Gradient",         "hand-derived backward()",    "automatic
    (GradientTape)", "automatic (autograd)"],
    ["Optimizer",        "S.Adam (written out)",       "keras.optimizers.Adam",
    "torch.optim.Adam"],
    ["Training loop",    "explicit for-loop",          "model.fit()",
    "explicit for-loop"],
    ["Evaluation",       "predict_classes + sklearn",   "predict + sklearn",
    "forward + sklearn"],
    ], columns=["Component", "Scratch", "TensorFlow/Keras", "PyTorch"])

display(component_table)

```

	Component	Scratch	TensorFlow/Keras	PyTorch
0	Data loading	ass4_utils -> NumPy arrays	same arrays	same arrays
1	Preprocessing	ass4_utils (shared)	shared	shared
2	Model definition	S.Sequential([...])	keras.Sequential([...])	nn.Module + nn.Sequential
3	Dense layer	S.Dense (x@W+b by hand)	keras.layers.Dense	nn.Linear
4	Activation	S.ReLU (mask multiply)	activation='relu'	nn.ReLU
5	Regularisation	S.Dropout (inverted)	keras.layers.Dropout	nn.Dropout
6	Loss	S.SoftmaxCrossEntropy	SparseCategoricalCrossentropy	nn.CrossEntropyLoss
7	Gradient	hand-derived backward()	automatic (GradientTape)	automatic (autograd)
8	Optimizer	S.Adam (written out)	keras.optimizers.Adam	torch.optim.Adam
9	Training loop	explicit for-loop	model.fit()	explicit for-loop
10	Evaluation	predict_classes + sklearn	predict + sklearn	forward + sklearn

```
In [20]: # Persist for 04_compare.ipynb
os.makedirs("results", exist_ok=True)
with open("results/01_diabetes130.json", "w") as f:
    json.dump({"meta": {k: v for k, v in meta.items() if k != "feature_names"},
               "results": [r.__dict__ for r in results],
               "component_table": component_table.to_dict(orient="records")},
              f, indent=2, default=str)
print("saved results/01_diabetes130.json")
```

saved results/01_diabetes130.json

Saving the trained models

The JSON above records what the models *scored*. It does not record what they *are* — rerunning any of these numbers would mean retraining from scratch.

`U.save_model` writes each leg in its framework's native format, under `results/models/<notebook>/`, with a JSON sidecar naming the architecture and the score the weights actually produced:

leg	file	reloads alone?
Scratch (NumPy)	<code>.npz</code> , one array per layer parameter	no
TensorFlow/Keras	<code>.keras</code> , graph + weights	yes
PyTorch	<code>.pt</code> , <code>state_dict</code>	no

`.pt` and `.npz` hold weights only, so reloading them means rebuilding the architecture first and passing the fresh instance to `U.load_model(..., model=...)`. That is the same contract `variants.py` uses for `results/variant_cache/`.

```
In [21]: # Weights are the model; metrics only describe it. Persist all three legs.
TAG = "01_diabetes130"

by_fw = {r.framework: r for r in results}

for obj, fw, stem in [
    (scratch_model, "Scratch (NumPy)", "mlp_scratch"),
    (keras_model, "TensorFlow/Keras", "mlp_keras"),
    (torch_model, "PyTorch", "mlp_torch"),
]:
    r = by_fw[fw]
    U.save_model(obj, f"{stem}", TAG, architecture=r.model, dataset=r.dataset,
                 epochs=r.epochs, test_accuracy=r.test_accuracy,
                 f1_macro=r.f1_macro)

display(U.list_models(TAG)[["name", "framework", "n_params", "dataset",
                             "test_accuracy"]])
```

```
saved results\models\01_diabetes130\mlp_scratch.npz (31,491 params, 116 KB)
saved results\models\01_diabetes130\mlp_keras.keras (31,491 params, 396 KB)
saved results\models\01_diabetes130\mlp_torch.pt (31,491 params, 126 KB)
```

	name	framework	n_params	dataset	test_accuracy
0	mlp_keras	keras	31491	Diabetes 130-US hospitals	0.607689
1	mlp_scratch	scratch	31491	Diabetes 130-US hospitals	0.609404
2	mlp_torch	torch	31491	Diabetes 130-US hospitals	0.610547

7. What this dataset shows

The three implementations agree. All three build the same 31,491-parameter function, and all three land within noise of each other on accuracy and macro F1. Different names — `S.Dense` , `keras.layers.Dense` , `nn.Linear` — do not imply different machine-learning concepts. The abstraction level changed; the mathematics did not.

Accuracy is the wrong headline here. With a 59 / 32 / 9 class split, the always-predict-`NO` baseline is already close to what the trained networks reach. The macro F1 column is what separates a model that learned something from one that learned the prior. Hospital readmission is genuinely hard to predict from administrative fields — the published literature on this dataset reports AUCs around 0.65, and nothing here contradicts that.

The tabular case is the control. No convolution appears anywhere in this notebook, because there is no spatial neighbourhood for a kernel to exploit. That is what makes the next two notebooks meaningful: when MNIST and CIFAR-10 introduce `Conv2D` , it will be a response to a property the data actually has, not a default choice.

The GPU did not help here — and that is worth pausing on. On this dataset the PyTorch leg is the *slowest* of the three despite running on an RTX 3060, while the hand-written NumPy leg is the fastest. The model is tiny and the data is tabular, so each batch carries too little arithmetic to amortise the cost of moving it to the device and back; the transfer overhead dominates. Hardware acceleration is not free, and it is not automatic. The two image notebooks provide exactly the condition it needs — enough work per batch to pay for itself — and there the ordering reverses sharply.

What the scratch leg bought. It is the only one whose gradients had to be proved correct rather than inherited from autodiff. In exchange, every quantity in the training cycle is visible — and the finite-difference check above is the evidence that "visible" also means "right".

Next: `02_mnist.ipynb` , where the data has spatial structure and the architecture changes to match it.